1. **Project Title: Analyzing Global Video Game Sales: A Look at Player Preferences Across Genres and Platforms**

Description: This project involves exploring a CSV dataset containing global video game sales information. Clean and analyze the data to identify the top-selling video game genres, platforms (e.g., PC, console, mobile), and regional trends (e.g., North America, Europe, Asia). Visualize your findings using charts and graphs to understand player preferences across different markets.

Guidelines:

- Clean the data for inconsistencies and missing values.
- Identify and analyze key metrics like sales figures, release dates, genres, and platforms.
- Segment the data by region and compare sales trends across different geographical areas.
- Use visualizations like bar charts, pie charts, and heatmaps to present your findings.

2. Project Title: Predicting Movie Success: Building a Decision Tree Model for Box Office Glory

Description: This project combines data analysis and machine learning. Utilize a CSV dataset containing movie information (e.g., genre, cast, budget, director) and box office performance. Clean and prepare the data for modeling. Build a decision tree classification model that predicts a movie's potential box office success based on its features.

Guidelines:

- Clean the data for inconsistencies and missing values.
- Preprocess the data for machine learning by handling categorical features (e.g., genre) and scaling numerical features (e.g., budget).
- Train a decision tree model on the prepared data, splitting it into training and testing sets.
- Evaluate the model's performance using metrics like accuracy and confusion matrix.
- Analyze the decision tree to understand the key factors influencing the model's predictions for movie success.

Project 3: Employee Attrition Analysis

- **Description:** Explore employee data to understand factors affecting attrition.
- **Objectives:**
    - Determine key factors contributing to employee attrition.

    - Visualize demographic distributions of employees.
    - Analyze the relationship between job roles and attrition.
- **Guidelines:**
    - Clean and preprocess the data.
    - Use bar charts and pie charts to visualize demographic distributions.
    - Create heatmaps or bar charts to show the relationship between job roles and attrition rates.

Project 4: Movie Ratings Analysis

- **Description:** Perform analysis on movie ratings data to understand viewer preferences and trends.
- **Objectives:**
    - Identify top-rated movies and genres.
    - Analyze rating distributions.
    - Find trends in viewer preferences over time.
- **Guidelines:**
    - Clean and preprocess the data.
    - Use bar charts to display top-rated movies and genres.
    - Use histograms or box plots to show rating distributions.
    - Perform trend analysis to identify changes in viewer preferences over time.

Project 5: E-commerce Order Analysis

- **Description:** Examine e-commerce order data to find purchasing trends and customer behavior.
- **Objectives:**
    - Identify peak order times.
    - Analyze customer purchasing patterns.
    - Determine top-selling products.
- **Guidelines:**
    - Clean and preprocess the data.
    - Use time series analysis to identify peak order times.
    - Use bar charts and scatter plots to analyze customer purchasing patterns.
    - Display top-selling products using bar charts.

Project 6: Health and Fitness Data

- **Description:** Explore health and fitness data to find trends in activity levels and health metrics.
- **Objectives:**
    - Analyze trends in activity levels over time.
    - Identify correlations between different health metrics.

    - Visualize the distribution of health metrics.
- **Guidelines:**
    - Clean and preprocess the data.
    - Use line charts to show trends in activity levels.
    - Use scatter plots or correlation matrices to analyze health metrics.
    - Create histograms or box plots to visualize health metric distributions.

Project 7: Titanic Survival Prediction

- **Description:** Analyze Titanic passenger data and build a model to predict survival.
- **Objectives:**
    - Perform exploratory data analysis (EDA) on passenger data.
    - Clean and preprocess the data for modeling.
    - Build and evaluate a classification model to predict survival.
- **Guidelines:**
    - Perform EDA to understand the data (visualize demographics, survival rates).
    - Handle missing values and encode categorical variables.
    - Use logistic regression or decision tree classifiers to build the model.
    - Evaluate the model using accuracy, precision, recall, and F1 score.

Project 8: Heart Disease Prediction

- **Description:** Analyze patient data and build a model to predict heart disease.
- **Objectives:**
    - Explore patient data and identify key health indicators.
    - Clean and preprocess the data for modeling.
    - Build and evaluate a classification model to predict heart disease.
- **Guidelines:**
    - Perform EDA to explore patient data (visualize distributions, correlations).
    - Handle missing values and scale numeric features.
    - Use decision tree or logistic regression classifiers to build the model.
    - Evaluate the model using metrics like accuracy, precision, recall, and F1 score.

Project 9: Credit Card Fraud Detection

- **Description:** Analyze credit card transaction data and build a model to detect fraud.
- **Objectives:**
    - Explore transaction data and identify fraudulent patterns.
    - Clean and preprocess the data for modeling.
    - Build and evaluate a classification model to detect fraud.
- **Guidelines:**
    - Perform EDA to explore transaction data (visualize distributions, correlations).
    - Handle missing values and encode categorical variables.

    - Use random forest or logistic regression classifiers to build the model.
    - Evaluate the model using metrics like precision, recall, and F1 score.

Project 10: Customer Churn Prediction

- **Description:** Analyze customer data and build a model to predict churn.
- **Objectives:**
    - Explore customer data and identify factors influencing churn.
    - Clean and preprocess the data for modeling.
    - Build and evaluate a classification model to predict churn.
- **Guidelines:**
    - Perform EDA to explore customer data (visualize distributions, correlations).
    - Handle missing values and encode categorical variables.
    - Use logistic regression or decision tree classifiers to build the model.
    - Evaluate the model using metrics like ROC-AUC, accuracy, and confusion matrix.

Project 11: Wine Quality Prediction

- **Description:** Analyze wine data and build a model to predict wine quality.
- **Objectives:**
    - Explore wine characteristics and identify factors affecting quality.
    - Clean and preprocess the data for modeling.
    - Build and evaluate a classification model to predict wine quality.
- **Guidelines:**
    - Perform EDA to explore wine data (visualize distributions, correlations).
    - Handle missing values and scale numeric features.
    - Use logistic regression or decision tree classifiers to build the model.
    - Evaluate the model using metrics like accuracy, precision, recall, and F1 score.

Project 12: Marketing Campaign Effectiveness

- **Description:** Analyze marketing campaign data and build a model to predict campaign success.
- **Objectives:**
    - Explore campaign data and identify factors influencing success.
    - Clean and preprocess the data for modeling.
    - Build and evaluate a classification model to predict campaign success.
- **Guidelines:**
    - Perform EDA to explore campaign data (visualize distributions, correlations).
    - Handle missing values and encode categorical variables.
    - Use logistic regression or random forest classifiers to build the model.
    - Evaluate the model using metrics like ROC-AUC, accuracy, and confusion matrix.